

The diagram shows a light blue shaded circle representing a ball B . A black vector v originates from the center of the circle. A red vector ξ also originates from the center. A blue arc on the boundary of the circle indicates the projection of ξ onto v , labeled $P_v \xi$. A right-angle symbol is shown at the tip of this projection. A dashed red circle is concentric with the blue circle, and a label $|v|$ is placed near the bottom of the dashed circle, indicating its radius.

HMC: $|v'|^2 = |v|^2 + \Delta t^2 |P_v \xi|^2$

RT: $|v'| = |v|$

The diagram illustrates a central body (light blue circle) with two concentric orbits. The inner orbit is a solid blue circle, and the outer orbit is a dashed red circle. A black dot marks a point on the blue orbit at the top. A red curve starts at this black dot, moves left and slightly up, then curves back down and left, ending at a red dot on the dashed red orbit. This red curve represents a perturbation or a transition from the blue orbit to the red orbit.

$$\mathbb{E}|v_N|^2 = |v_0|^2 + ND\sigma^2\Delta t^2$$

$|v_N| = |v_0|$, $\hat{v}_N$ **diffuses**